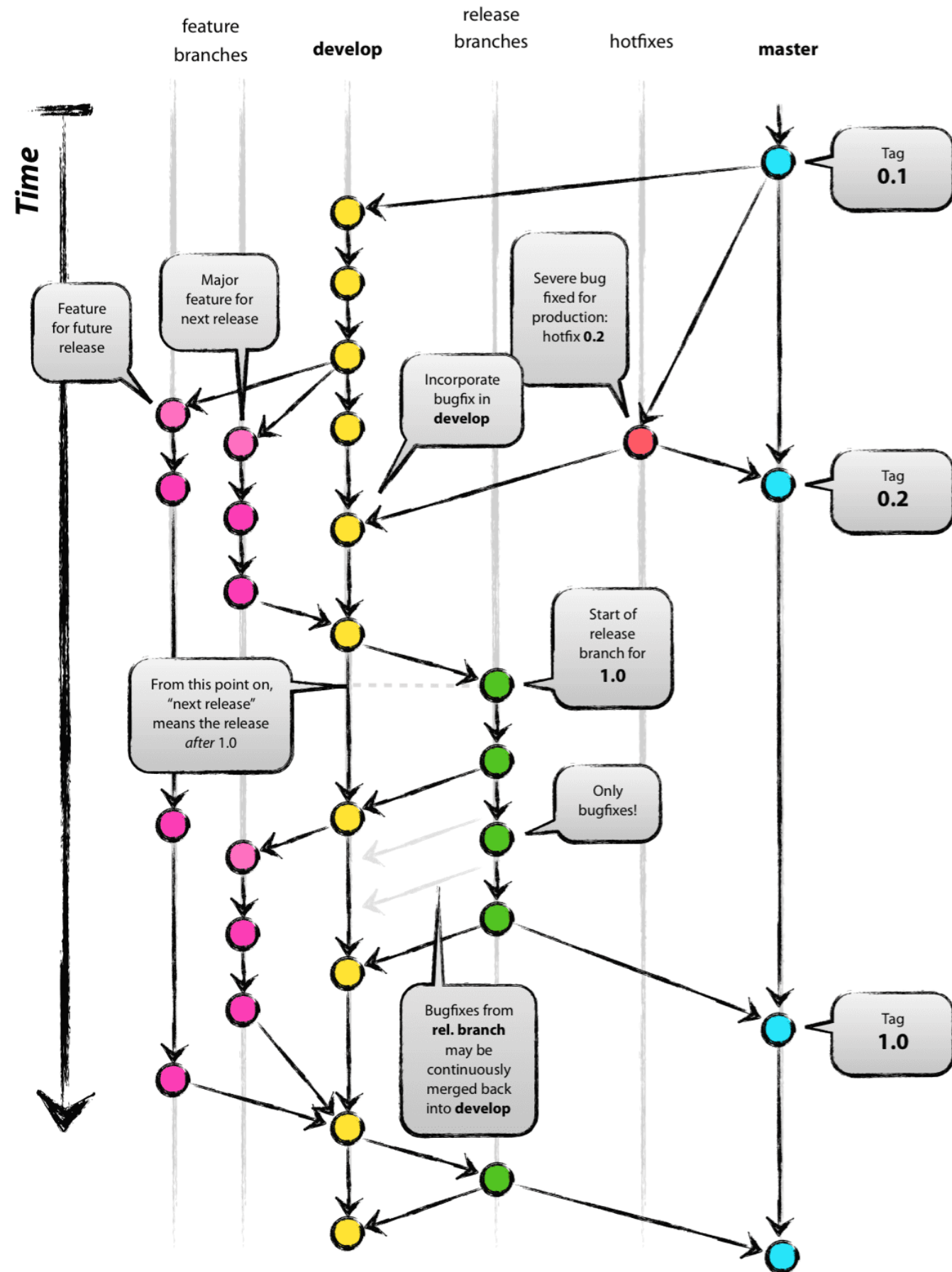


VCS & Project Management – Git & GitLab

Xiang Li @ UCchip

Agenda

- Git
 - Git Basics & Concepts
 - Git Usage
 - Git Workflow
- GitLab
 - GitLab Topics
 - Scrum & GitLab
 - Issue Management
 - Markdown

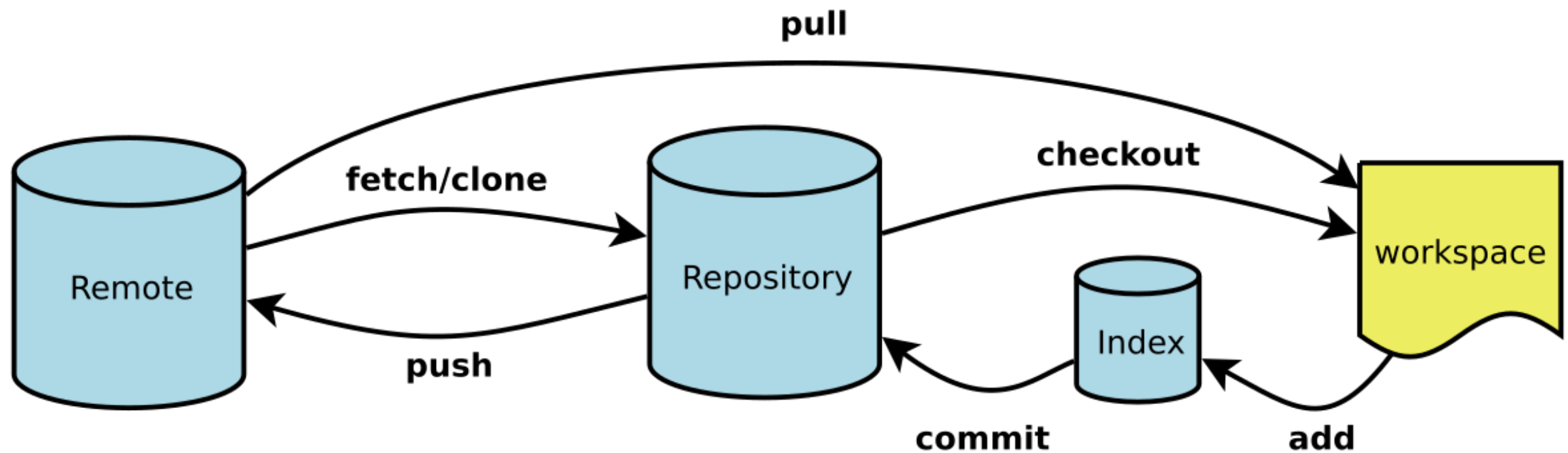


Git Basics

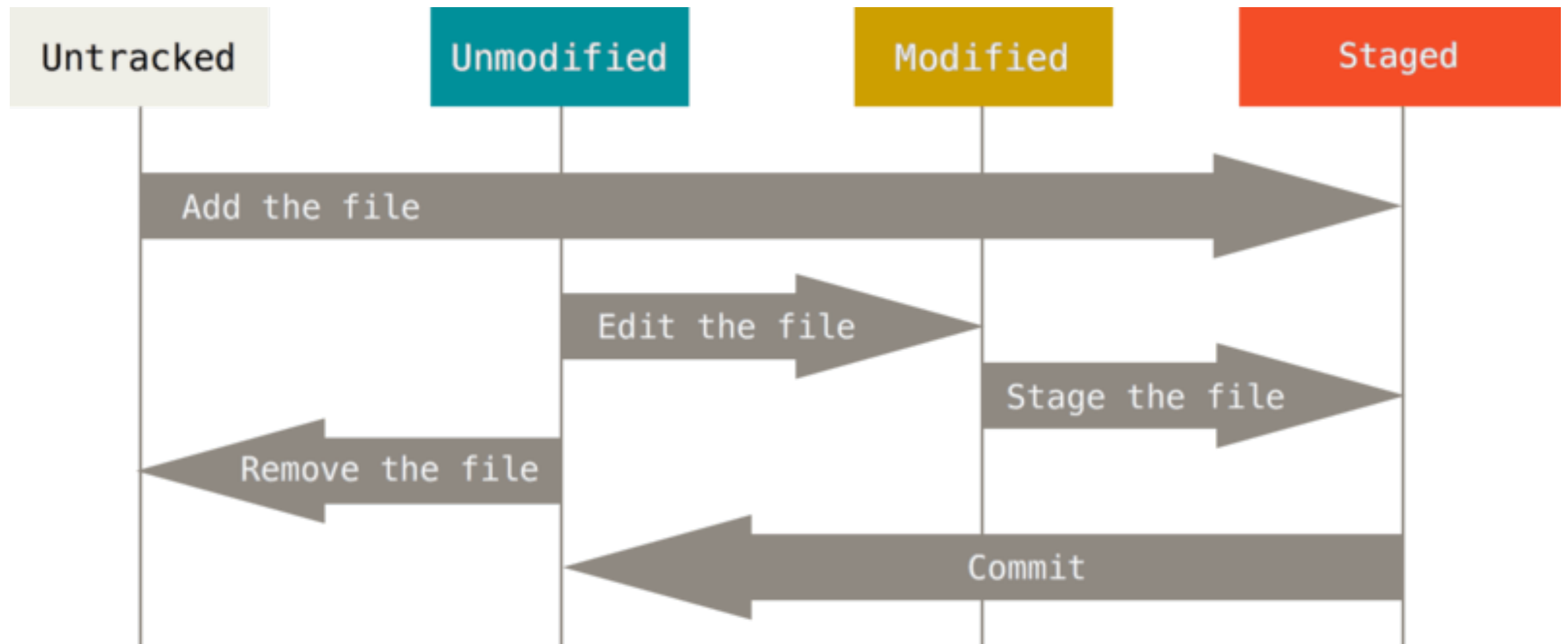
GitHub Git Cheat Sheet

Three Local Areas

- Repository / History
- Index / Stage
- Workspace / Working Directory



File Life Cycle



A Visual Git Reference

Ref & Refspec

Git References - *A pointer to a commit*

- heads (branches)
- remotes (remote branches)
- tags

```
git show-ref  
git ls-remote origin
```

Git Refspecs - *Reference Specification*

```
+src:dst  
.git/config  
git push origin HEAD:master
```

git branch upstream

```
# View branch upstream  
git branch -vv
```

```
# Set the upstream of the <branch-name>  
git branch -u <remote-name>/<remote-branch-name> <branch-name>
```

```
# Push and track upstream  
git push <remote-name> <local-branch-name>:<remote-branch-name>
```

git add/checkout -p

Interactive operation

Add interactively
git add -p

Check out interactively
git checkout -p

git log

```
git config --global alias.l  
    "log --all --graph --pretty='format:%C(auto)%h%d %s    %Cblue%an, %ar'"
```

```
git blame
```

```
git log -L 360,+30:register.py
```

Pretty Formats

git revert

Revert a commit; Generate a new one

```
git revert <commit>
```

Revert a merge

```
git log <commit>
```

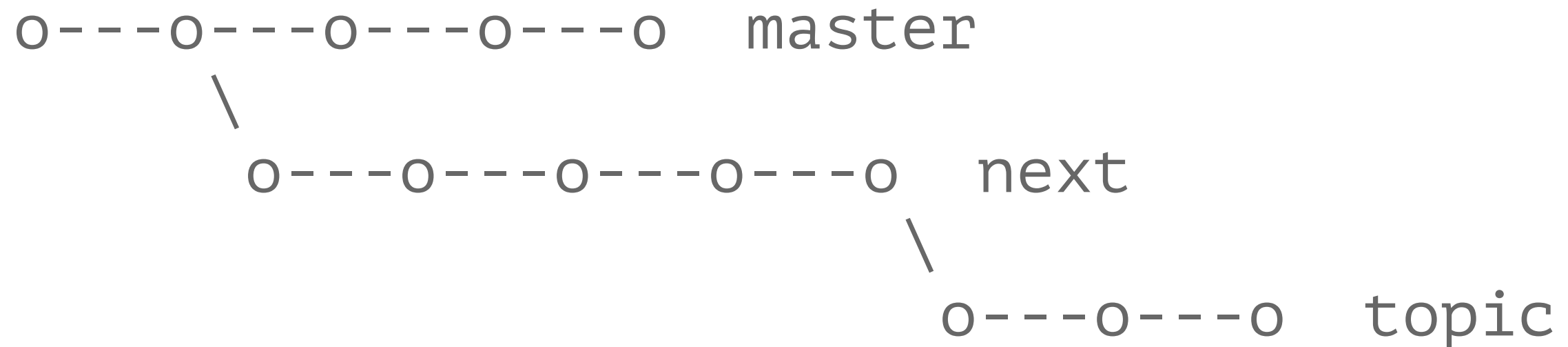
Restore to the state as in 1

```
git revert <commit> -m 1
```

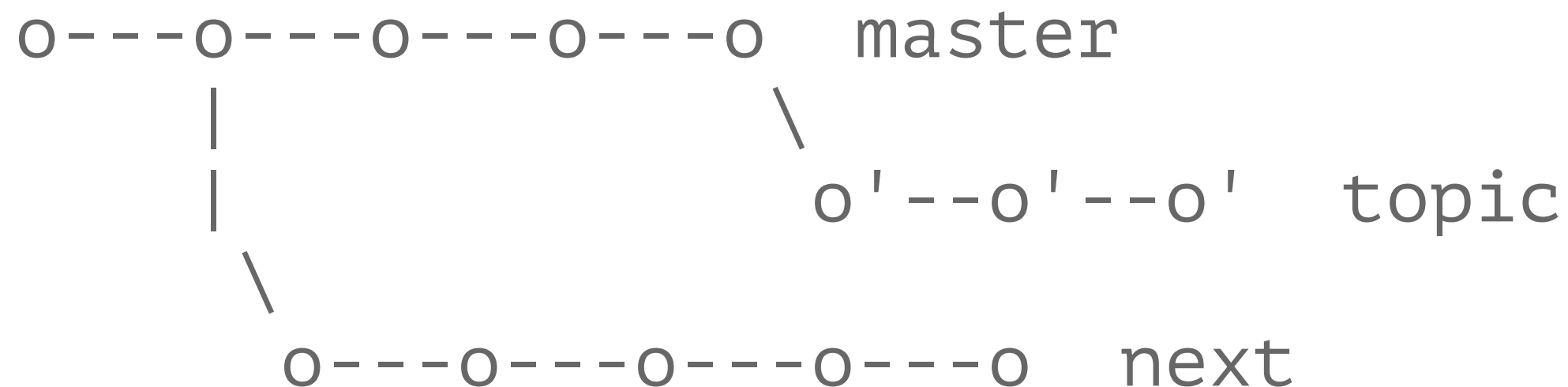
Restore to the state as in 2

```
git revert <commit> -m 2
```

git rebase



`git rebase --onto master next topic`



git cherry-pick

Apply the changes introduced by some existing commits

Usage

- Pick a commit from a feature branch as a hotfix

```
git cherry-pick <commit>
```

```
# Add 'cherry picked from commit'
```

```
git cherry-pick -x
```

```
git cherry-pick --no-commit
```

Git Tutorials & References

GitHub Git Cheat Sheet

Atlassian Git

Official Book

Official Reference

A Visual Git Reference

A bunch of good git tutorials

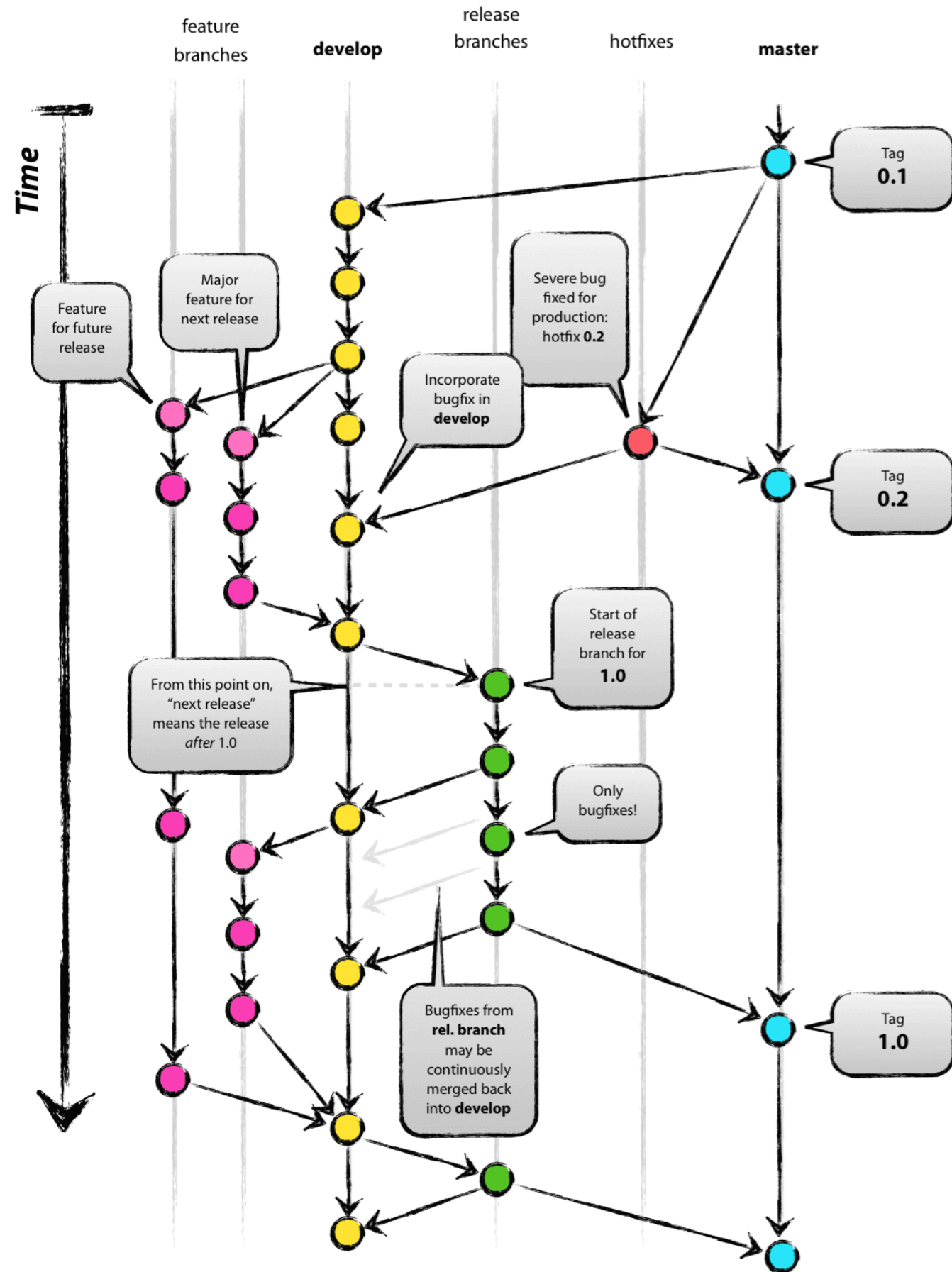
Git Workflows

- Centralized
- Feature Branch
- Forking
 - Repo per developer
 - For separation
- Git Flow^{Driessen, 2010}

Driessen, 2010 [A successful Git branching model](#)

Gitflow Workflow

- master
 - Protected branch
 - Only merge release
- develop
 - Automatic tests
- feature
 - Branch from develop
 - Merge into develop
- release
 - Branch from develop
 - Merge into develop and master
- hotfix
 - Branch from master
 - Merge into develop and master
- Notes
 - `merge --no-ff`
 - Remove finished branch



Q & A

GitLab

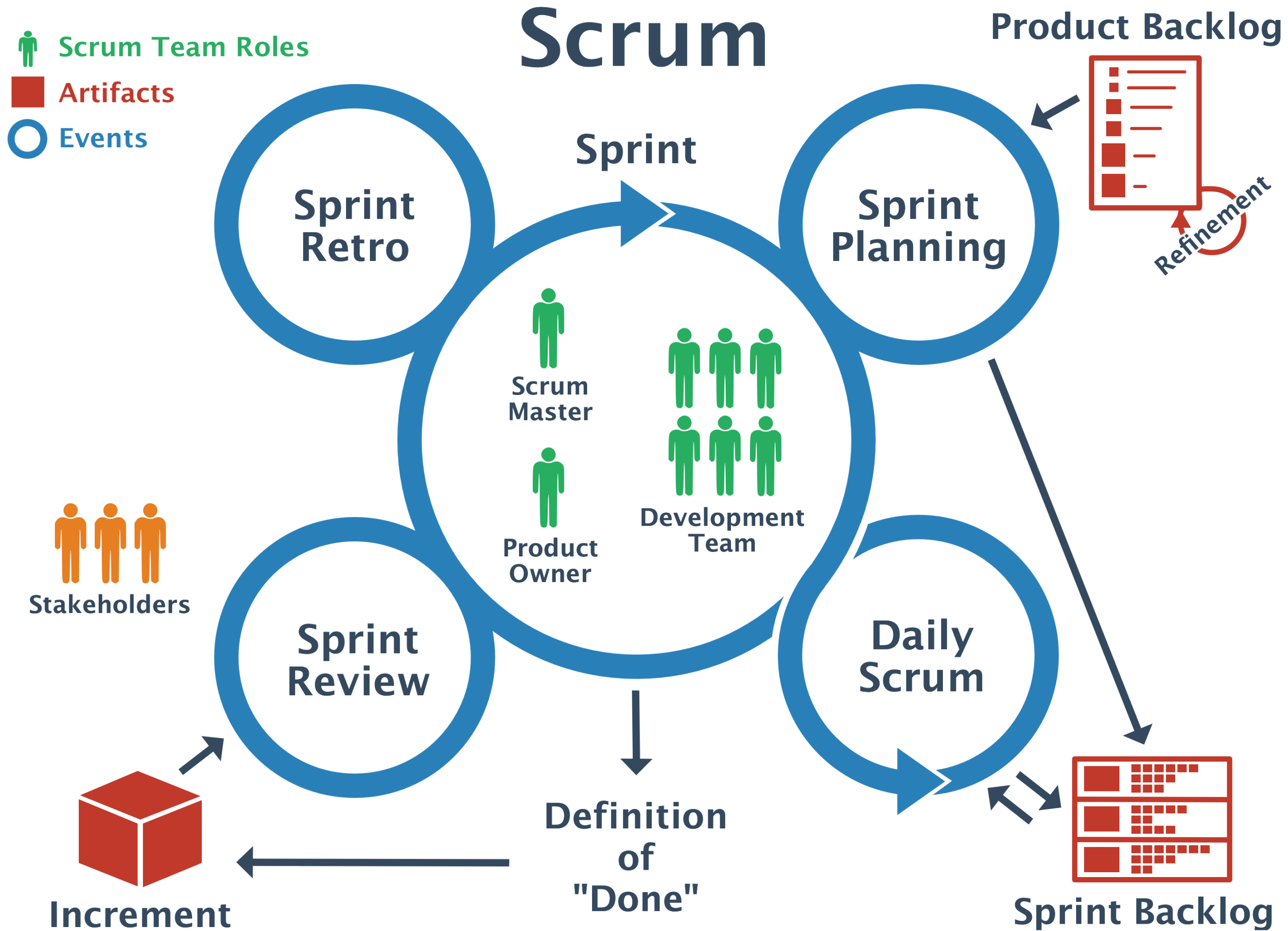
*A combination of
version control and
project management*

GitLab Topics

- Group: *group member*
- Project: *project member*
 - Repository
 - Issues
 - Labels
 - Board
 - Milestones
 - Wiki
 - Snippets
 - CI/CD
- Notifications



GitLab



Scrum References

Scrum.org

Scrum Alliance

Scrum Guides

Atlassian Scrum Tutorial

Mapping Scrum to GitLab

Scrum	GitLab
Sprints	Milestones
User stories	Issues
Tasks	Tasks
Epic	Epic (EE only)
Points and estimation	Issue weights (EE only)
Product backlog	Issue lists and labels
Kanban board	Issue board

GitLab Issues

Your issue is the Single Source of Truth

Use Cases:

- Discussing the implementation of a new idea
- Tracking tasks and work status
- Accepting feature proposals, questions, support requests, or bug reports
- Elaborating on new code implementations

GitLab Issues

Content

- Title
- Description and tasks
- Comments and other activity

People

- Author (Owner)
- Assignee

State

- Status: Open/Closed
- Life cycle: Labels

Planning and Tracking

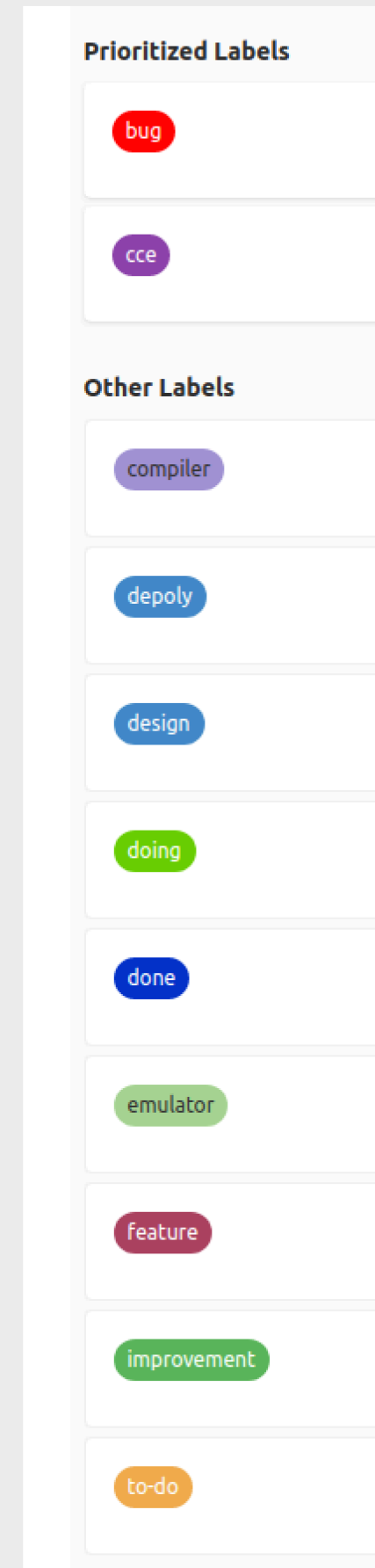
- Milestone
- Time tracking
- Due date
- Labels

Others

- Related issues (Through crosslinking)

Manage Issue Labels

- Development cycle
 - Open (Unassigned)
 - To-do
 - Design
 - Implementation
 - Review
 - Test
 - Deploy
 - Closed
- Development category
 - Bug
 - New feature
 - Improvement
- Functional category
 - CCE
 - CCE Compiler
 - CCE Emulator



Issue Tips

- Manage issue life cycle through labels
- Referencing what's related
 - References in commits messages
 - References in descriptions and comments
 - Every commit belongs to an issue
- Code review
- @Mention others and assign (Notification in header)
- Show comments only
- Quick actions¹
- Container issues

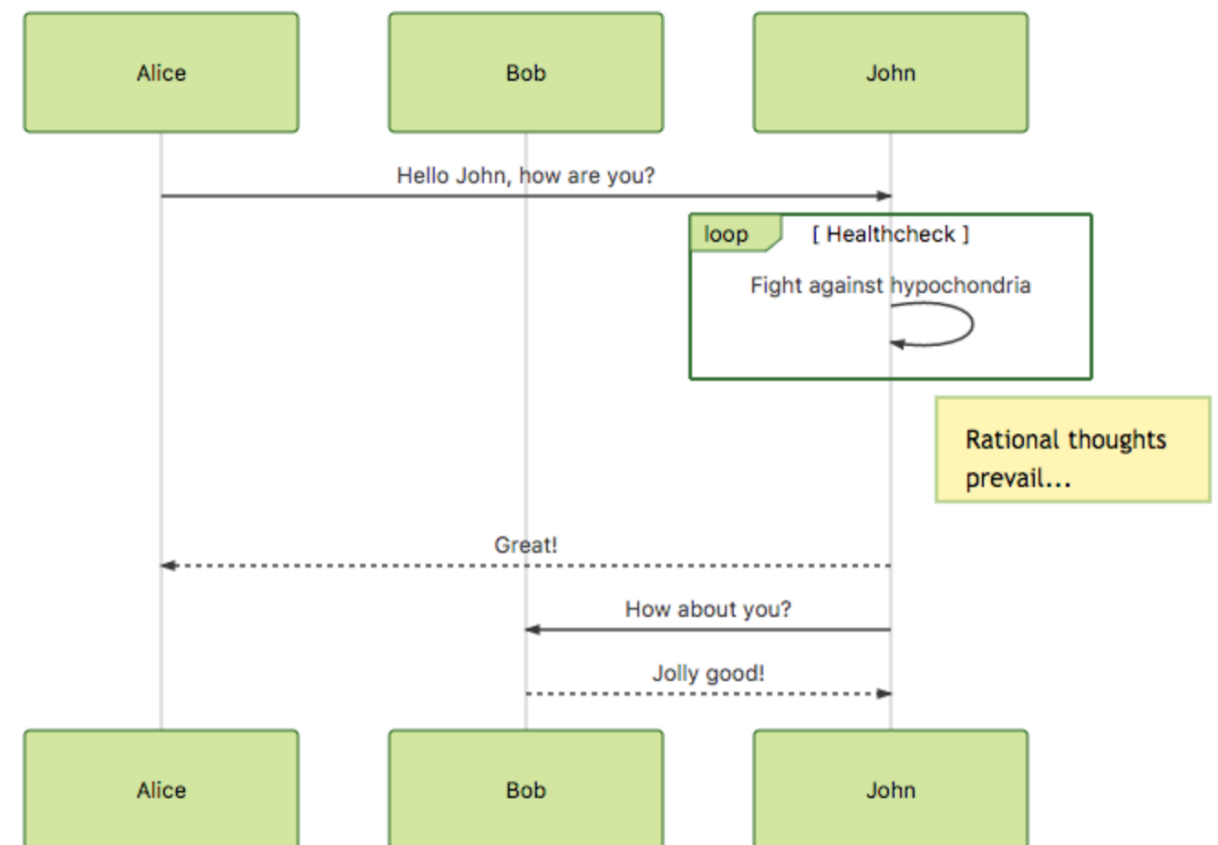
¹ Quick actions

Example: CCE Tools

Markdown Syntax

- Markdown Basic Syntax
 - Headings
 - Emphasis
 - Lists
 - Links
 - Images
- Markdown Extended Syntax
 - Code blocks
 - Tables
 - Tasks
- GitLab Flavored Markdown (GFM)
 - Diagrams (**Mermaid**): Flowchart, Sequence diagram, Gantt diagram
 - Math (**KaTeX**)
- Special GitLab references

```
sequenceDiagram
    participant Alice
    participant Bob
    Alice->>John: Hello John, how are you?
    loop Healthcheck
        John->>John: Fight against hypochondria
    end
    Note right of John: Rational thoughts <br/>prevail!
    John-->>Alice: Great!
    John->>Bob: How about you?
    Bob-->>John: Jolly good!
```



Wiki

Wiki is also a repo

- Full path in filename as folders
- Relative path image
- Docs can also be put into the repo
- PDF viewer in repo

Q & A



Thank you!

Tutorials & References

Git

[GitHub Git Cheat Sheet](#)

[Atlassian Git](#)

[Official Book](#)

[Official Reference](#)

[A Visual Git Reference](#)

[A bunch of good git tutorials](#)

Scrum

[Scrum.org](#)

[Scrum Alliance](#)

[Scrum Guides](#)

[Atlassian Scrum Tutorial](#)

Markdown Syntax

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